Roadmap

What is implemented, what was left out of the MVP, the technical debt, and what a further month would add.

Implemented

Requirements R1-R6 from the case, as closed by the code:

Requirement	State	Where
R1 source exchange: PCAP/PCAPNG, sessions, directions, reassembly, provenance, diagnostics	done	`src/capture/`
R2 research tools: search, comparison, hex view, action journal	done	`src/capture/`, `src/ui/`
R3 interpretations: declarative rules, versioning, reuse on new data	done	`src/protocol/`, `src/hypothesis/`
R4 verification: corpus, counterexamples, visible applicability, recheck	done	`src/hypothesis/`
R5 portable project: relative paths, manifest, versions, results	done	`src/project/`
R6 reuse: report (Markdown/HTML), machine-readable result (JSON), CLI	done	`src/report/`, `src/*/cli.py`

Out of MVP scope

The extensions the case names that were not built, with the reason and the shape a first version would take.

Extension	Why left out	First version
Explainable pattern proposals: suggest field layouts the researcher did not state	Needs a search over offsets, widths and readings that is a project on its own; the current tool scores readings the researcher supplies	Enumerate candidate field boundaries over the framed bytes and rank by support/contradict, reusing the existing scoring
Automatic request/response linking	Needs the response rule, which was not derived; the corpus carries both directions but only the request direction is interpreted	Match by transaction id or by timestamp window, then check the field correspondence
Behaviour comparison between captures	The corpus is one protocol; a second protocol would be needed to make the comparison meaningful	Diff two projects by rule version and by message counts
Complex structures: nested, variable-length, repeated fields	The rule schema is flat fields; nesting would change the schema	A field kind `list` with a count field and a child layout
Controlled experiments: change one byte and observe	No live target; the captures are recorded	A project action that rewrites a message and re-runs the rule

Technical debt

Item	Impact	Path
encrypted payloads	out of scope by design	none; the tool reports the bytes it sees
memory scales with the capture	all packets held at once	session-by-session parse with provenance streamed to disk (see `docs/BENCHMARK.md`)
formatting and typing deferred	`ruff format --check` and `mypy` still report findings in modules outside the audited zone (the extended `ruff` set itself is enabled)	re-enable `ruff format --check .` and `mypy .` one zone at a time (see `docs/AUDIT.md`)

Product plan for a further month

1. Derive the response rule, then link requests to responses by transaction id; the corpus already holds both directions.
2. Pattern proposal: search field boundaries and rank readings, so a researcher starts from suggestions instead of a blank rule.
3. Second protocol in the corpus, so behaviour comparison has something to compare and the applicability boundary is exercised on real variance.
4. Memory-bounded mode by default: parse and reassemble session by session with provenance written as it is produced.
5. Fragment reassembly (IPv4 and IPv6) in the capture layer.